

Topology-Preserving Transform (TPT) State-Variable Filters

Zero-Delay Feedback Filter Discretization

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Abstract

This paper details the continuous-time and discretized equations of the 2-pole active State Variable Filter (`[filter~]` and `[svfilter~]`) implemented using Vadim Zavalishin's Topology-Preserving Transform (TPT) method.

1 Continuous-Time Active SVF Equations

The continuous-time state-variable filter integrates high-pass y_{HP} , band-pass y_{BP} , and low-pass y_{LP} outputs:

$$y_{HP}(t) = x(t) - R \cdot y_{BP}(t) - y_{LP}(t) \quad (1)$$

$$y_{BP}(t) = \omega_0 \int y_{HP}(t) dt \quad (2)$$

$$y_{LP}(t) = \omega_0 \int y_{BP}(t) dt \quad (3)$$

where $\omega_0 = 2\pi f_c$ is the cutoff angular frequency and $R = \frac{1}{Q}$ is the damping factor.

2 Bilinear Transform Discretization & TPT Structure

Applying the trapezoidal integration rule with pre-warped gain $g = \tan\left(\frac{\pi f_c}{f_s}\right)$ and resonance damping $k = \frac{1}{Q}$:

$$v_H(n) = \frac{x(n) - (g + k)s_1(n-1) - s_2(n-1)}{1 + g(g + k)} \quad (4)$$

$$v_B(n) = g \cdot v_H(n) + s_1(n-1) \quad (5)$$

$$v_L(n) = g \cdot v_B(n) + s_2(n-1) \quad (6)$$

State registers $s_1(n)$ and $s_2(n)$ are updated at each sample:

$$s_1(n) = 2v_B(n) - s_1(n-1) \quad (7)$$

$$s_2(n) = 2v_L(n) - s_2(n-1) \quad (8)$$